

# Towards Robustness : Adversarial Attack on Bangla Language Models

SM Musfique Rahman, Mohammad Rabbu, Ziaul Hassan, MD Rafsanul Islam Nelay, Sheikh Fardeen Ishaque, G.

M. Shahariar Shibli, Sajib Kumar Saha Joy

Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Email: {musfique0608, md.rabbu2441, ziaulhasanf, rafsanulnay}@gmail.com

**Abstract**—Adversarial robustness in natural language processing (NLP) has been widely studied for English and other high-resource languages, but remains largely unexplored for Bangla, one of the most spoken yet underrepresented languages. In this paper, we present the first systematic study of adversarial attacks and defenses in Bangla text classification. We compile five benchmark datasets spanning diverse tasks—sentiment analysis, clickbait detection, sarcasm detection, fake news detection, and hate speech classification—and evaluate the vulnerability of pre-trained deep learning classifiers under five representative attack strategies: paraphrasing, gradient-based perturbations, homoglyph substitution, random masking, and important-token masking. Our experiments demonstrate that Bangla classifiers are highly sensitive to even minimal perturbations, with gradient-based and homoglyph attacks being particularly effective across tasks. To mitigate this, we propose a lightweight defense based on *perplexity thresholding*, which reliably detects adversarial samples with competitive  $F_2$ , precision, and recall across datasets. This work provides the first empirical foundation for understanding adversarial threats in Bangla NLP and highlights directions for building more robust systems in low-resource languages.

**Index Terms**—adversarial attack, adversarial defense, Bangla NLP, text classification, perplexity thresholding, robustness, low-resource language

## I. Introduction

Despite remarkable progress in natural language processing (NLP), neural text classifiers remain highly vulnerable to adversarial perturbations[1]–[3]. Such perturbations, although often small and semantically preserving, can drastically change model predictions and raise serious concerns about fairness, trustworthiness, and robustness in real-world applications. Prior work has investigated a variety of attack strategies, including character-level edits, synonym substitutions[1], gradient-based manipulations[4], [5], and paraphrasing[6], mostly targeting English and other high-resource languages. However, adversarial robustness of Bangla NLP models remains largely unexplored, despite Bangla being the seventh most spoken language worldwide[7], [8]. This gap is particularly critical as Bangla NLP is increasingly being used in sensitive applications such as sentiment and emotion analysis[9], [10], fake news detection [11], hate speech detection[12], sarcasm identification[13], and clickbait detection[14] where adversarial vulnerabilities may lead to harmful consequences. Therefore, it is imperative to systematically study adversarial attacks and defenses in the Bangla context to ensure robustness and reliability.

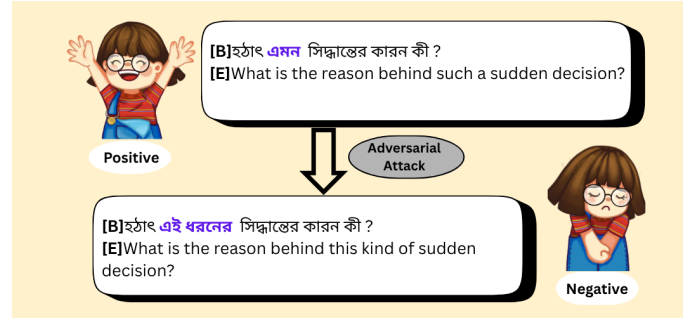


Figure 1: Adversarial examples in which a slight change to the input text alters the prediction. Here **B** represents the Bangla text and **E** represents the English text.

This raises several key questions: Are Bangla text classification models similarly vulnerable to adversarial attacks as their English counterparts? How do different attack strategies (e.g., paraphrasing vs. gradient-based vs. character-level manipulations) compare across diverse Bangla tasks? Can lightweight detection methods be designed to distinguish adversarially perturbed Bangla text from clean inputs?

In this paper, we attempt to answer these questions through a systematic empirical study. Specifically, we investigate five representative Bangla text classification tasks sentiment analysis, clickbait detection, sarcasm detection, fake news detection, and hate speech classification covering a diverse range of linguistic and semantic phenomena. We evaluate five attack strategies, spanning both black-box and white-box settings, and benchmark their effectiveness in inducing misclassification.

Our main contributions are summarized as follows:

- We conduct the first systematic study of adversarial attacks in Bangla NLP across five benchmark text classification datasets, highlighting the vulnerability of models in a low-resource setting.
- We implement and evaluate five attack strategies—paraphrase, gradient-based, homoglyph substitution, random masking, and important-token masking—providing a comparative analysis of their effectiveness.
- We propose a lightweight defense mechanism based on perplexity thresholding, which effectively detects adversarial samples across datasets with competitive precision and recall.

Our study sheds light on the adversarial landscape of Bangla NLP and establishes a foundation for future robustness research in low-resource languages.

## II. Related Works

The vulnerability of neural language models to adversarial attacks has emerged as a critical concern in natural language processing (NLP), as these attacks can severely compromise model reliability, fairness, and linguistic integrity [15], [16]. A broad range of research has been conducted to evaluate adversarial robustness and develop effective attack strategies for both encoder-based and decoder-based architectures. These methods span across character-level, word-level, and phrase-level perturbations, often employing synonym substitutions [1], masked token replacements [17]–[19], or gradient-based token optimization [4], [5].

Jin et al. [1] introduced TextFooler, a synonym-substitution-based method that generates adversarial examples by identifying important tokens and replacing them with semantically similar alternatives. Their approach substantially degraded model accuracy across multiple benchmark datasets and architectures. Similarly, Ren et al. [20] proposed Probability Weighted Word Saliency (PWWS), which scores token importance and applies strategic synonym replacements. These word-level attacks maintain fluency while successfully misleading deep classifiers such as WordCNN, LSTM, and BERT.

Character-level approaches such as TextBugger [21] and DeepWordBug [22] modify textual input at the sub-word level via insertions, deletions, and character substitutions. Though effective in low-resource settings, these perturbations often result in reduced readability. To enhance attack realism, masked language model (MLM) guided methods such as BERT-Attack [19] and BAE [17] utilize masked token predictions from pre-trained transformers to construct adversarial variants with minimal semantic shift.

Wallace et al. [4] introduced universal adversarial triggers—short, input-agnostic token sequences optimized via gradients to consistently induce misclassification across NLP tasks. Guo et al. [5] proposed a general-purpose gradient-based attack that optimizes over a distribution of adversarial examples using continuous-valued perturbations. Both methods reveal the susceptibility of transformer-based models to gradient-driven manipulations, even under minimal input modification.

Recent work has also extended adversarial analysis to **Large Language Models (LLMs)**. Zhu et al. [23] proposed PromptBench to evaluate prompt-level vulnerabilities in models like ChatGPT, Vicuna, and LLaMA. Their findings highlight that even advanced models are susceptible to simple word or sentence-level manipulations, which can drastically affect downstream task performance. These insights are particularly important as LLMs are increasingly adopted in real-world systems without adequate robustness checks.

To the best of our knowledge, no prior work has explored adversarial attacks targeting Bangla language models. Given that Bangla is a low-resource language with limited benchmark datasets and pretrained models, this area remains significantly under-explored. Our work aims to bridge this gap by systematically evaluating the vulnerability of Bangla NLP models to a diverse set of adversarial attack strategies, providing a foundational analysis for future robustness research in Bangla language processing.

## III. Problem Formulation

Let  $\mathcal{X} = \{x_1, x_2, \dots, x_N\}$  be a collection of Bangla text samples and  $\mathcal{Y} = \{y_1, y_2, \dots, y_N\}$  the corresponding label set. Given a pre-trained classification model  $F : \mathcal{X} \rightarrow \mathcal{Y}$ , the goal of the adversary is to generate an adversarial example  $x^{adv}$  for a clean input  $x \in \mathcal{X}$ , such that the perturbed text remains semantically similar to  $x$ , but causes the model to predict an incorrect label, i.e.,

$$F(x^{adv}) \neq F(x)$$

This work explores multiple attack strategies under this setting, aiming to expose the vulnerability of Bangla text classifiers by inducing misclassification through minimal and meaning-preserving perturbations.

## IV. Attack Design

### A. Datasets

We evaluate adversarial vulnerabilities across five Bangla text-classification datasets covering diverse NLP tasks: sentiment analysis, fake news detection, sarcasm detection, clickbait detection, and hate speech classification.

SentNoB is a sentiment analysis dataset consisting of public comments from social media platforms, spanning 13 domains such as politics, education, and agriculture. Each comment is labeled as positive, negative, or neutral.

Bangla-English Fake News includes 14,537 news articles labeled as real (label-1) or fake (label-0). The dataset spans multiple news categories, with 37% from national news, 13% international, and the remainder from various other domains.

BanglaSARC is a sarcasm detection dataset containing 5,112 Bangla-language comments and status updates collected from Facebook, YouTube, and blogs. Each entry is labeled as sarcastic or non-sarcastic.

BanglaBait is a clickbait detection dataset comprising 15,406 labeled and 65,405 unlabeled news articles sourced from clickbait-heavy Bangla news portals. Each article includes a title, body, and metadata, with expert annotations.

BD-SHS is a large-scale hate speech dataset with over 50,200 offensive comments gathered from social networking sites. It is significantly larger than previously available Bangla hate speech corpora and serves as a benchmark for toxic content detection.

## B. Target Models

For white-box attacks, we evaluated five task-specific Bangla classification models based on pre-trained BERT and MuRIL architectures. These include publicly available models fine-tuned on **sentiment analysis**<sup>1</sup>, **fake news detection**<sup>2</sup>, **sarcasm detection**<sup>3</sup>, **clickbait detection**<sup>4</sup>, and **hate speech detection**<sup>5</sup> tasks. The models were obtained from Hugging-Face and correspond to the five downstream tasks used in this study.

Each model was evaluated using the test split of its respective dataset. We do not perform any additional training or fine-tuning. Each model is evaluated in a white-box setting using the test split of its respective dataset.

For black-box attacks, we evaluated three multilingual instruction-tuned large language models (LLMs) accessed via the Unsloth library in 4-bit quantized format: **LLaMA 3.2-3B Instruct**<sup>6</sup>, **Qwen2.5-7B Instruct**<sup>7</sup>, and **Mistral 7B Instruct v0.3**<sup>8</sup>. These models were not fine-tuned on any of the downstream tasks and were not fine-tuned on any of the downstream classification tasks.

These LLMs were accessed in an instruction-following setup and evaluated as-is without access to model weights or architecture internals, conforming to a strict black-box evaluation scenario.

## C. Attack Methods

We evaluate five different attack methods used in the experiments.

**Paraphrasing:** This method[6] generates semantically equivalent but lexically altered versions of the original text using a Bangla paraphrasing model. The goal is to preserve the original meaning while modifying surface-level syntax, vocabulary, or phrasing. We employ a pre-trained transformer-based Bangla paraphraser[24] to rephrase input texts. If the model’s prediction changes due to the paraphrased input, the attack is considered successful. This approach simulates natural linguistic variability and tests whether models rely too heavily on specific lexical patterns.

**Random Word Masking:** In this method, one word is randomly selected and masked from each sentence in the input text. The selected word is either replaced with a special token or a plausible substitute. Using the masking-based perturbation strategy introduced in [18], we adopt the Replace action for our experiments.

Let  $X = [x_1, x_2, \dots, x_n]$  be the input sequence and  $\text{pos}$  be the randomly selected word position in a sentence. The perturbed input  $X'$  is constructed as:

$$X' = [x_1, \dots, x_{\text{pos}-1}, [\text{MASK}], x_{\text{pos}+1}, \dots, x_n] \quad (1)$$

To generate a filled version of the masked input using a language model, we define:

$$\text{fill\_mask}(X', \text{pos}) = [x_1, \dots, x_{\text{pos}-1}, T[0], x_{\text{pos}+1}, \dots, x_n] \quad (2)$$

where  $T[0]$  is the top predicted token from the language model used to fill the masked position. The goal is to introduce localized perturbations that minimally affect human understanding while disrupting the model’s internal representation.

**Important Word Masking:** This attack targets the most influential word in each sentence by identifying tokens that have a high impact on the model’s prediction [19]. The word importance is determined using the Adv-OLM method [25], which combines occlusion-based scoring and language model predictions to rank token relevance. The top-ranked word is then masked or replaced, preserving the sentence’s fluency while perturbing the model’s decision boundary.

Let  $x = [x_1, x_2, \dots, x_n]$  be the input sequence and  $f_y(x)$  denote the model’s confidence score (e.g., probability or logit) for the true or predicted class  $y$ . To estimate the importance of each word  $x_i$ , we compute:

$$r_i = \left| f_y(x) - f_y(x^{\setminus i}) \right| \quad (3)$$

where  $x^{\setminus i}$  represents the input with the  $i$ -th word masked or removed. The word with the highest score  $r_i$  is selected as the most influential token in the sentence.

**Homoglyph Attack:** Homoglyph[26] attack method substitutes characters in the input text with visually similar Unicode counterparts, known as homoglyphs. For example, a Bangla or Latin character may be replaced with a nearly identical-looking variant from another script or character set. These perturbations are imperceptible to human readers but can confuse tokenizers or embedding layers, leading to misclassifications. The attack simulates subtle, stealthy manipulations in text and is effective even under minimal modification constraints.

**Gradient-Based Attack:** Gradient-Based [5] attack method appends a sequence of specially crafted tokens, known as universal adversarial triggers, to the input text. These triggers are learned in a preprocessing phase by optimizing the token embeddings using gradient signals from the model. Specifically, gradients of the classification loss with respect to the input are used to iteratively refine a fixed set of tokens that can induce misclassification across a wide range of inputs. Once generated, the same trigger sequence is appended to every test sample during inference. The original input text and model parameters remain unchanged.

<sup>1</sup><https://huggingface.co/armansakif/bengali-fake-news>

<sup>2</sup><https://huggingface.co/ka05ar/banglabert-sentiment>

<sup>3</sup><https://huggingface.co/raquiba/sarcasm-detection-BanglaSARC>

<sup>4</sup>[https://huggingface.co/samanjoy2/banglaclibert\\_base](https://huggingface.co/samanjoy2/banglaclibert_base)

<sup>5</sup><https://huggingface.co/Hate-speech-CNERG/bengaliabusive-MuRIL>

<sup>6</sup><https://huggingface.co/unsloth/Llama-3.2-3B-Instruct>

<sup>7</sup><https://huggingface.co/unsloth/Qwen2.5-7B-Instruct>

<sup>8</sup><https://huggingface.co/unsloth/mistral-7b-instruct-v0.3-bnb-4bit>

Let  $T = [t_1, t_2, \dots, t_k]$  denote the trigger token sequence appended to each input  $x$ . The tokens are optimized to maximize the loss  $\mathcal{L}(f(x \oplus T), y)$ , where  $\oplus$  represents concatenation. The trigger update at iteration  $i$  is computed as:

$$T^{(i+1)} = T^{(i)} + \eta \cdot \nabla_T \mathcal{L}(f(x \oplus T^{(i)}), y) \quad (4)$$

where  $\eta$  is the learning rate, and  $\nabla_T \mathcal{L}$  denotes the gradient of the loss with respect to the trigger tokens. Only the trigger tokens are updated — the model  $f$  remains fixed throughout.

#### D. Experimental Setup & Hyper-parameters

We evaluate adversarial robustness across five Bangla text classification tasks: sentiment analysis, fake news detection, sarcasm detection, clickbait detection, and hate speech classification. For each task, we use pretrained transformer-based models from HuggingFace that were already fine-tuned on the corresponding dataset. We perform no additional training, tuning, or modification of these models.

Adversarial examples are generated using five attack methods: paraphrasing, random word masking, homoglyph substitution, important word masking, and a gradient-based trigger attack. The first four methods are applied without any parameter tuning or search; they operate directly on the input text using rule-based or sampling-based perturbations.

The gradient-based attack is implemented using a universal trigger approach, where a fixed sequence of tokens is learned through gradient-based optimization. These trigger tokens are generated in a separate preprocessing step by identifying patterns that consistently cause the model to misclassify inputs. Once generated, the same triggers are appended to all test inputs during inference. The model parameters remain unchanged throughout the process.

For black-box evaluation, we employ three instruction-tuned LLMs — LLaMA 3.2–3B Instruct, Qwen2.5–7B Instruct, and Mistral 7B Instruct v0.3 — accessed through the Unsloth library in 4-bit quantized format. These models are evaluated in a zero-shot setting using manually constructed Bangla prompts that contain either the original or adversarial text.

#### E. Evaluation Metrics

To evaluate the effectiveness of adversarial attacks across all models and tasks, we use the following two metrics:

- **Accuracy:** Measures the proportion of correctly classified samples in the dataset:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(f(x_i) = y_i)$$

where  $x_i$  is an input sample,  $y_i$  is its corresponding ground truth label, and  $\mathbb{I}$  is the indicator function.

- **Attack Success Rate (ASR):** The proportion of originally correct predictions that are flipped after applying the adversarial perturbation. It is defined as:

$$\text{ASR} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(f(x_i) \neq f(x_i^{adv}))$$

where  $x_i$  is a clean input,  $x_i^{adv}$  is its adversarial version, and  $\mathbb{I}$  is the indicator function. A higher ASR indicates greater model vulnerability.

For white-box models, predictions are obtained directly from the classification head. For black-box LLMs, outputs are interpreted from zero-shot prompt completions using keyword matching or rule-based label extraction.

#### F. Attack Performance & Result

Table I summarizes the accuracy of the baseline classification models and the attack success rates (ASR) of five adversarial strategies—Gradient Attack, Homoglyph Substitution, Random Word Masking, Important Word Masking, and Paraphrasing—across five benchmark datasets. Gradient-based attacks achieved the highest ASR in most cases, indicating their strong capability to generate effective universal triggers. For example, in the Sentiment and Hate Speech classification tasks, the ASR reaches 99.56% and 96.56%, respectively. However, these attacks do not preserve the semantics or naturalness of inputs, making them less stealthy.

Among the input-specific perturbation methods, Homoglyph and Paraphrasing attacks generally perform better than masking-based methods, especially on datasets like Clickbait and Sarcasm. Important Word Masking is particularly effective in semantic-rich tasks like Sarcasm and Sentiment, where the model heavily depends on a small set of influential tokens. In contrast, Random Word Masking yields lower success rates across most datasets, suggesting that disrupting less critical words has limited impact on model predictions.

Table II presents the ASR of four adversarial attacks on three instruction-tuned large language models (LLaMA, Qwen, and Mistral) across the same datasets. Overall, the LLMs exhibit varying degrees of robustness depending on the attack type and task. Mistral shows stronger resilience in Clickbait and Hate Speech detection, while Qwen is notably more vulnerable in tasks like Sarcasm and Sentiment. Important Word Masking and Paraphrasing remain consistently effective across models, reaffirming that perturbing semantically critical parts of the input can substantially degrade LLM performance.

These results highlight that while white-box attacks like Gradient-based methods offer high success rates, black-box and input-specific perturbations remain viable and interpretable strategies for evaluating model robustness in real-world applications.

#### G. Error Analysis

Although adversarial attacks achieved moderate to high success rates across different tasks, a significant portion of adversarial examples failed to fool the models. To better understand these failure cases, we manually inspected samples from multiple attacks and datasets. Table III presents representative examples of unsuccessful adversarial attempts, along with the original and perturbed texts, predictions, and observations. We highlight the perturbed tokens in **bold**.

| Dataset    | Accuracy | Attack Success Rate (ASR) |           |                     |                        |            |
|------------|----------|---------------------------|-----------|---------------------|------------------------|------------|
|            |          | Gradient Attack           | Homoglyph | Random Word Masking | Important Word Masking | Paraphrase |
| Clickbait  | 65.72    | 35.28                     | 76.92     | 11.04               | 8.77                   | 8.00       |
| Fakenews   | 74.55    | 91.90                     | 23.61     | 55.91               | 57.61                  | 89.19      |
| Hatespeech | 87.89    | 96.56                     | 6.24      | 5.53                | 40.11                  | 11.14      |
| Sarcasm    | 98.88    | 98.81                     | 31.96     | 2.13                | 38.56                  | 18.92      |
| Sentiment  | 16.02    | 99.56                     | 19.05     | 9.02                | 34.74                  | 21.56      |

Table I: Success rates (%) of five adversarial attack strategies—Gradient Attack, Homoglyph Attack, Random Word Masking, Important Word Masking, and Paraphrase Attack—evaluated across five benchmark datasets. Higher success rates indicate greater susceptibility of the model to the corresponding attack.

| Dataset    | Homoglyph |       |         | Important Word Masking |       |         | Random Word Masking |       |         | Paraphrase |       |         |
|------------|-----------|-------|---------|------------------------|-------|---------|---------------------|-------|---------|------------|-------|---------|
|            | Llama     | Qwen  | Mistral | Llama                  | Qwen  | Mistral | Llama               | Qwen  | Mistral | Llama      | Qwen  | Mistral |
| Clickbait  | 27.76     | 10.74 | 1.64    | 26.73                  | 3.91  | 0.27    | 25.92               | 4.15  | 0.47    | 29.08      | 5.01  | 2.06    |
| Fakenews   | 19.17     | 6.38  | 27.88   | 19.78                  | 2.70  | 14.67   | 15.74               | 3.20  | 21.85   | 26.04      | 5.16  | 30.80   |
| Hatespeech | 2.94      | 20.66 | 16.96   | 2.83                   | 16.15 | 15.28   | 2.74                | 13.10 | 13.44   | 2.74       | 21.59 | 21.67   |
| Sarcasm    | 21.44     | 19.19 | 14.46   | 21.57                  | 40.52 | 13.77   | 17.51               | 11.66 | 10.52   | 26.21      | 19.27 | 17.55   |
| Sentiment  | 13.49     | 23.90 | 23.01   | 13.68                  | 24.91 | 25.85   | 9.46                | 17.21 | 22.89   | 18.60      | 25.03 | 32.28   |

Table II: Attack success rates (%) of four adversarial techniques—Homoglyph, Important Word Masking, Random Word Masking, and Paraphrasing—on five benchmark datasets (Clickbait, Fake News, Hate Speech, Sarcasm, and Sentiment) across three LLMs (LLaMA, Qwen, and Mistral). Higher values indicate greater model vulnerability.

## H. Human Evaluation

We manually assessed adversarial quality from a human perspective by sampling **20 pairs per attack** (pooled across all datasets: Sentiment, Sarcasm, Hate Speech, Fake News, Clickbait). For each original–adversarial pair we judged: (i) *Grammaticality* (1–5), (ii) *Semantic similarity* (0–1), (iii) *Human label consistency* (whether a human would still assign the same task label; %), and (iv) *Readability* issues (percentage marked hard-to-read due to homoglyph/character noise). Results are in Table IV.

| Attack                 | Grammar | Similarity | Human Acc. | Hard-to-Read |
|------------------------|---------|------------|------------|--------------|
| Paraphrase             | 3.00    | 0.25       | 85%        | 60%          |
| Random Masking         | 3.60    | 0.45       | 90%        | 25%          |
| Important Word Masking | 2.75    | 0.55       | 85%        | 60%          |
| Homoglyph              | 2.95    | 0.53       | 85%        | 60%          |
| Gradient (GA)          | 2.50    | 0.38       | 90%        | 70%          |

Table IV: Human-style evaluation per attack (20 random pairs per attack pooled across all datasets). Grammar: 1–5; Similarity: 0–1; Human Acc.: human-assigned label consistency; Hard-to-Read: fraction flagged as difficult due to visual/character noise.

Overall, adversarial edits moderately reduce grammatical fluency (Grammar  $\approx$  2.5–3.6). Paraphrase often preserves label salience for humans (85% consistency) but shows lower perceived similarity when content is heavily rephrased. Random Masking remains most readable (25% hard-to-read) with small semantic drift, whereas Homoglyph and GA edits degrade readability (60–70%) and similarity more noticeably, yet humans still recover the correct label in most cases (85–90%).

## V. Defense Mechanism

### A. Method

Our defense mechanism is based on the observation that adversarially perturbed text often exhibits a noticeable increase in perplexity when evaluated by a well-trained language model. Previous works have explored perplexity-based detection in various NLP settings, including spam filtering, machine translation robustness, and adversarial example detection[27]–[30], showing that abnormal perplexity patterns can effectively separate natural from manipulated text. We utilize *perplexity thresholding* [27] to distinguish clean inputs from adversarial ones.

| Attack Type            | Original Text                       | Adversarial Text                    | True Label | Pred. Before | Pred. After | Observation                                                                                                           |
|------------------------|-------------------------------------|-------------------------------------|------------|--------------|-------------|-----------------------------------------------------------------------------------------------------------------------|
| Masking (Sentiment)    | “এই সিনেমাটা দারুণ ভালো”            | “এই সিনেমাটা ... ভালো”              | Positive   | Positive     | Positive    | The core sentiment word (“ভালো”) remained unchanged, so the model correctly classified it despite masking.            |
| Homoglyph (Sarcasm)    | “কি চমৎকার আইডিয়া!”                | “কি চমৎকার আইদায়া!”                | Sarcastic  | Sarcastic    | Sarcastic   | Perturbation only altered one character visually, while semantic meaning stayed intact, leading to a failed attack.   |
| Paraphrase (Fake News) | “খালেদার জামিনের মেয়াদ বেড়েছে”    | “বেগম খালেদার জামিন বাড়ানো হয়েছে” | Fake       | Fake         | Fake        | Although paraphrased, the semantic meaning (extension of bail) remained identical, resulting in unchanged prediction. |
| Masking (Clickbait)    | “অবিশ্বাস্য! আপনি জানবেন না এই খবর” | “অবিশ্বাস্য! ... এই খবর”            | Clickbait  | Clickbait    | Clickbait   | The trigger word (“অবিশ্বাস্য”) was retained, which strongly signals clickbait, making the attack unsuccessful.       |

Table III: Representative failure cases of adversarial attacks across multiple datasets. Highlighted words indicate perturbed tokens. Despite perturbations, the semantic cues remained strong enough for the model to correctly classify the input.

Given an input sequence  $x = (w_1, w_2, \dots, w_T)$ , the perplexity is computed as:

$$\text{PPL}(x) = \exp \left( -\frac{1}{T} \sum_{t=1}^T \log P(w_t | w_{<t}) \right),$$

where  $P(w_t | w_{<t})$  is the predicted conditional probability of token  $w_t$  given its preceding tokens. Higher perplexity values indicate lower likelihood under the language model and may suggest unnatural or adversarial perturbations.

We transform the perplexity value into a binary feature by applying a decision threshold  $\tau$ , learned from validation data. Alongside sequence length, this feature is fed into a lightweight classifier to detect adversarial samples. This approach ensures that the defense remains model-agnostic and does not require access to the internal parameters of the target classifier.

### B. Experimental Setting & Evaluation Metrics

We evaluate the proposed perplexity-based defense on all five Bangla text classification datasets used in our adversarial attack experiments. The defense leverages *perplexity thresholding*, where the perplexity of each input text is computed using a pre-trained language model. Candidate thresholds are selected from a linear range between the minimum perplexity value and the 97<sup>th</sup> percentile of the distribution. For each threshold, we generate a binary feature indicating whether the input perplexity falls below it, and pair this with the input sequence length as features.

A LightGBM classifier is trained using a 60%/20%/20% train/validation/test split, optimizing the decision boundary to distinguish between clean and adversarial texts. The optimal perplexity threshold is determined through grid search over the candidate set, maximizing the  $F_2$ -score on the validation set to prioritize recall in detection.

We report  $F_2$ -score as the primary evaluation metric, alongside precision, recall, accuracy, and confusion matrix analysis. These metrics are computed separately for the validation and test sets to assess both the tuning effectiveness and the generalization capability of the proposed defense.

### C. Experimental Results

Tables V and VI summarize the detection performance of our perplexity-thresholding defense across five adversarial attacks (Paraphrase, Gradient, Homoglyph, Random Masking, Important Masking) and five Bangla text classification datasets (Sentiment, Clickbait, Sarcasm, Fake News, and Hate Speech). For each attack-dataset pair, we report the optimal perplexity threshold ( $\tau$ ) selected to maximize the  $F_2$ -score on the validation set, alongside  $F_2$ , Precision ( $P$ ), Recall ( $R$ ), and Accuracy (Acc) on the held-out test set.

Overall, Gradient-based attacks consistently achieved the highest  $F_2$ -scores across datasets, with values exceeding 0.94 in most cases. Homoglyph attacks also yielded strong detection scores, particularly on Sarcasm ( $F_2 = 0.932$ ) and Sentiment ( $F_2 = 0.931$ ). In contrast, Random Masking and Important Masking produced lower  $F_2$  values, reflecting

their subtle perturbation patterns that are harder to detect via perplexity measures.

### D. Analysis

A cross-metric inspection reveals several key observations:

- **Gradient attacks** are highly detectable, with both Precision and Recall often exceeding 0.90, suggesting that their perturbations introduce distributional shifts easily captured by perplexity-based filtering.
- **Paraphrase attacks** achieve moderate detection rates ( $F_2$  between 0.51 and 0.79), indicating that semantic-preserving rewording partially evades detection, though higher thresholds improve Recall.
- **Homoglyph attacks** perform comparably to Gradient attacks in several datasets, demonstrating that visual character substitutions significantly impact language model perplexity.
- **Random Masking** yields very high Recall on certain datasets (e.g.,  $R = 0.994$  for Sentiment) but low Precision, meaning the method flags many benign instances as adversarial.
- **Important Masking** shows mixed results, with reasonable Recall in some datasets (e.g.,  $R = 0.758$  for Sentiment) but overall lower Accuracy, suggesting room for improvement in balancing false positives and false negatives.

These findings confirm that perplexity thresholding is an effective defense for attacks that substantially disrupt token distribution, but less effective for subtle, meaning-preserving perturbations. Future work may combine this approach with semantic similarity checks or ensemble filtering for improved robustness.

## VI. Conclusion & Future Works

In this paper, we presented the first systematic exploration of adversarial robustness in Bangla text classification. By evaluating five diverse attack methods across five benchmark datasets, we demonstrated that Bangla NLP models are highly susceptible to both black-box and white-box perturbations. Among the methods, gradient-based attacks were the most effective, consistently reducing model accuracy to near-random levels. To mitigate these vulnerabilities, we introduced a perplexity thresholding defense, which effectively distinguished adversarial from clean samples while maintaining competitive detection performance.

Our findings underscore the urgent need for robustness evaluation in Bangla NLP, a low-resource but globally significant language. As future work, we plan to expand this study in several directions: (i) extending attacks to generative large language models (LLMs) in Bangla; (ii) exploring adversarial training and hybrid defenses to further improve resilience; and (iii) curating larger and more diverse Bangla adversarial benchmarks to support community-wide progress. We hope this work not only strengthens the safety of Bangla NLP systems but also inspires similar studies in other underrepresented languages.

| Method            | Sentiment |              | Clickbait |              | Sarcasm |              | Fake News |              | Hate Speech |              |
|-------------------|-----------|--------------|-----------|--------------|---------|--------------|-----------|--------------|-------------|--------------|
|                   | $\tau$    | $F_2$        | $\tau$    | $F_2$        | $\tau$  | $F_2$        | $\tau$    | $F_2$        | $\tau$      | $F_2$        |
| Paraphrase        | 10.13     | 0.788        | 7.17      | 0.362        | 10.026  | 0.759        | 6.675     | 0.515        | 10.639      | 0.703        |
| Gradient          | 9.275     | <b>0.969</b> | 5.14      | <b>0.773</b> | 7.327   | <b>0.991</b> | 5.039     | <b>0.940</b> | 13.88       | <b>0.981</b> |
| Homoglyph         | 12.29     | 0.931        | 10.02     | 0.788        | 11.448  | 0.932        | 8.789     | 0.869        | 11.923      | 0.895        |
| Random Masking    | 3.268     | 0.823        | 22.53     | 0.133        | 5.680   | 0.537        | 7.337     | 0.773        | 8.223       | 0.476        |
| Important Masking | 20.00     | 0.677        | 21.617    | 0.685        | 5.656   | 0.666        | 2.707     | 0.706        | 38.177      | 0.441        |

Table V: Per-attack detection performance across datasets:  $\tau$  and  $F_2$ . Best values per metric in **bold**.

| Method            | Sentiment    |              |              | Clickbait    |              |              | Sarcasm      |              |              | Fake News    |              |              | Hate Speech  |              |              |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   | $P$          | $R$          | Acc          | $P$          | $R$          | Acc          | $P$          | $R$          | Acc          | $P$          | $R$          | Acc          | $P$          | $R$          | Acc          |
| Paraphrase        | 0.627        | 0.842        | 0.679        | 0.572        | 0.332        | 0.546        | 0.554        | 0.836        | 0.592        | 0.525        | 0.513        | 0.525        | 0.626        | 0.725        | 0.646        |
| Gradient          | <b>0.927</b> | 0.981        | <b>0.953</b> | 0.538        | <b>0.867</b> | 0.578        | <b>0.958</b> | <b>0.999</b> | <b>0.977</b> | <b>0.792</b> | <b>0.986</b> | <b>0.865</b> | <b>0.913</b> | <b>1.000</b> | <b>0.951</b> |
| Homoglyph         | 0.828        | 0.961        | 0.884        | <b>0.686</b> | 0.819        | <b>0.723</b> | 0.840        | 0.958        | 0.890        | 0.663        | 0.943        | 0.732        | 0.715        | 0.955        | 0.787        |
| Random Masking    | 0.487        | <b>0.994</b> | 0.487        | 0.540        | 0.112        | 0.499        | 0.501        | 0.547        | 0.513        | 0.493        | 0.901        | 0.495        | 0.528        | 0.465        | 0.524        |
| Important Masking | 0.473        | 0.758        | 0.469        | 0.506        | 0.752        | 0.511        | 0.530        | 0.711        | 0.527        | 0.503        | 0.785        | 0.507        | 0.535        | 0.423        | 0.527        |

Table VI: Per-attack detection performance across datasets: Precision, Recall, and Accuracy. Best values per metric in **bold**.

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